

Motion Control

Objective

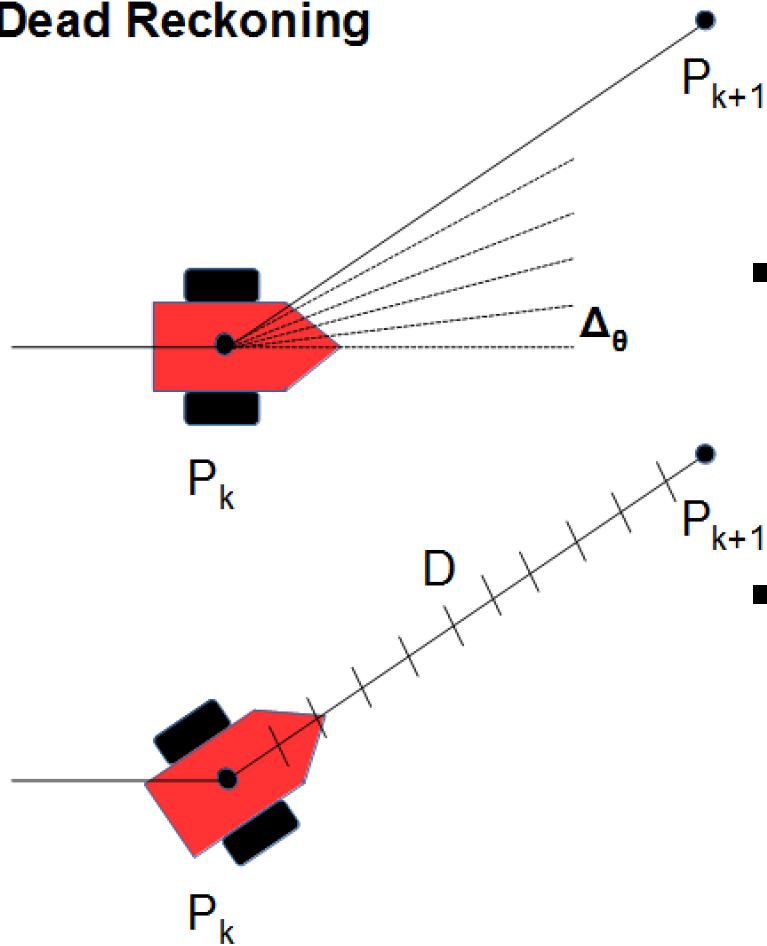
- Understanding the use of the kinematic models.
- Having some notions about the kinematic control of the mobile robot.

Use of the kinematic models

- Kinematic models are used both for location and motion control.
At this point we will illustrate how a robot can traverse a path using its kinematic model.
- A path is an ordered list of poses (x, y, θ) or positions (x, y) where:
 - the first element of the list is the pose / current position of the robot;
 - the last element of the list is the target that the robot wants to achieve.
- A trajectory can be approximated by line segments and circles.
- Along a path the robot moves from the current pose/position to the next one, or go through each segment one after the other.
- This can be done in an open loop or closed loop:
 - Open loop: the error (distance to the path) is not taken into account;
 - Closed loop: the error is corrected along the robot's movement.

Use of the kinematic models

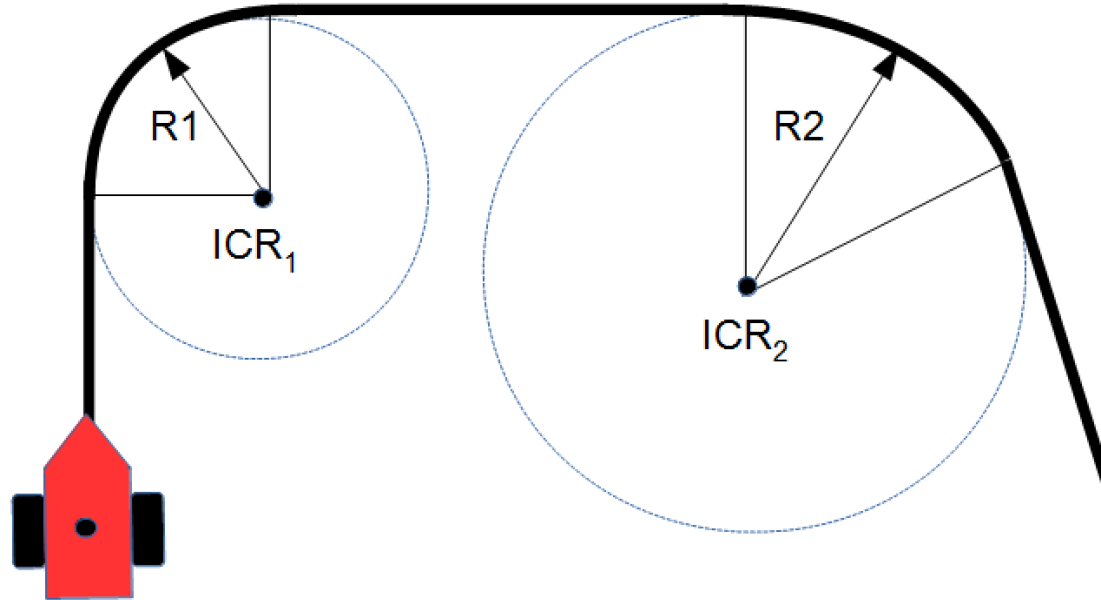
Dead Reckoning



- Initially the robot is reorienting toward the next point. This is done by applying opposed wheels speeds during the time needed to steer the robot.
- Once the robot is oriented toward its goal, velocities are applied on wheels for the necessary time to reach the distance D .
- For each iteration (Δt) the robot's pose is updated according to the incremental kinematic model until the target pose is reached.

Note: In practice controllers P / PI / PID are used to control the robot's targeted heading and location

Open-Loop Motion



- Following this trajectory, the robot can maintain a constant linear velocity.
- In the circular segments the robot adapts the difference of wheels velocities in order to track the desired curve (i.e. match the ICR with the center of the circle)
- To do so, the kinematic differential model is used to determine if the arc of a circle or line segment were covered in full.

Open-Loop Motion

The measured robot position is not fed back for velocity or position control.

It has several disadvantages:

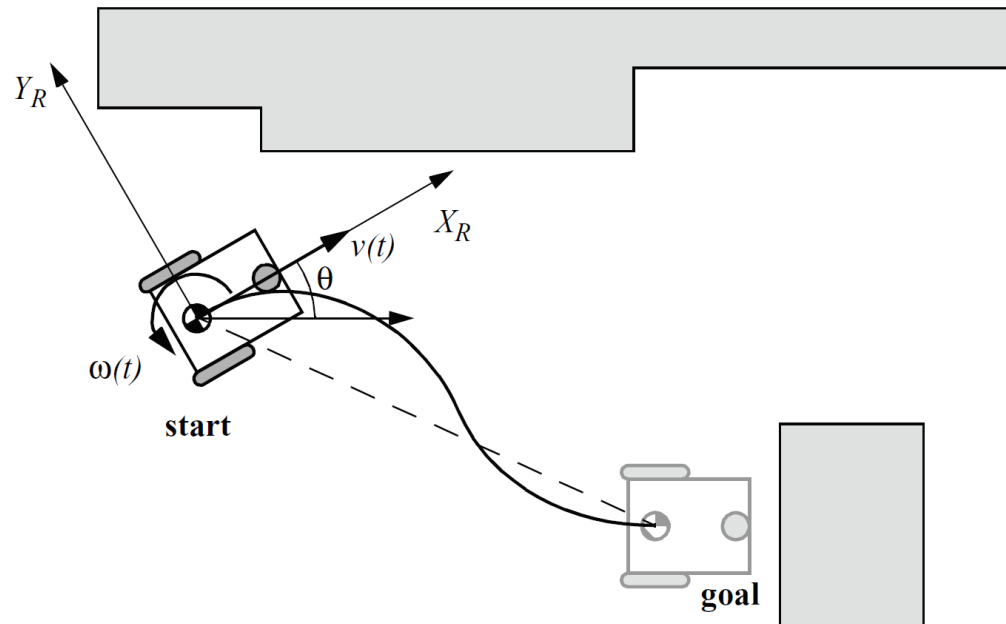
- It is not at all an easy task to pre-compute a feasible trajectory if all limitations and constraints of the robot's velocities and accelerations have to be considered.
- The robot will not automatically adapt or correct the trajectory if dynamic changes of the environment occur.
- The resulting trajectories are usually not smooth, as the transitions from one trajectory segment to another are, for most of the commonly used segments (e.g., lines and part of circles), not smooth => discontinuity in the robot's acceleration.

Position control

- Also called Feedback control
- Real-state feedback controller is a more appropriate approach in motion control of a mobile robot .
- With such a controller the robot's path-planning task is reduced to setting intermediate positions (sub-goals) lying on the requested path.

Position control: problem statement

- The robot has an arbitrary position and orientation and a predefined goal position/orientation
- The pose error vector in robot reference frame $\{X_R, Y_R, \theta\}$ is $e = {}^R[x, y, \theta]^T$ with x , y , and θ being the goal coordinates



Position control: problem statement

- The task of the controller is to find a control matrix K

$$K = \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \end{bmatrix} \text{ with } k_{ij} = k(t, e)$$

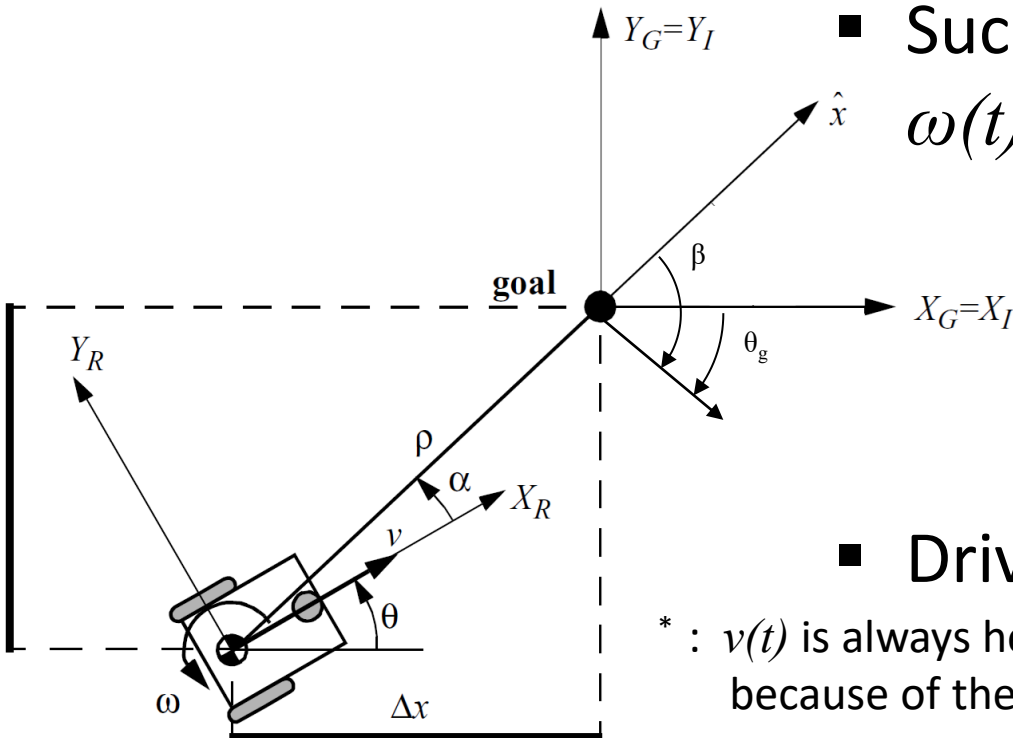
- Such that the control of $v(t)$ and $\omega(t)$

$$\begin{bmatrix} v(t) \\ \omega(t) \end{bmatrix} = K.e = K \begin{bmatrix} x \\ y \\ \theta \end{bmatrix}^R$$

$$\lim_{t \rightarrow \infty} e(t) = 0$$

- Drives the error e toward zero*

* : $v(t)$ is always heading in the X_R direction of the robot frame because of the non-holonomic constraint



Position control: kinematic model

- Let's assume that the goal is at the origine of the inertial frame
- Kinematic model of a differential drive robot in the inertial frame:

$${}^I \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \quad \text{with } \dot{x} \text{ and } \dot{y} \text{ the linear velocities in } X_I \text{ and } Y_I,$$

$$\text{if } \alpha \in I_1 \text{ with } I_1 = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

$$\rho = \sqrt{\Delta x^2 + \Delta y^2}$$

$$\text{transformation in polar coordinates : } \alpha = -\theta + \text{atan2}(\Delta y, \Delta x)$$

$$\beta = -\theta - \alpha$$

Position control: kinematic model

- The system description in the new polar coordinates:

$$\begin{bmatrix} \dot{\rho} \\ \dot{\alpha} \\ \dot{\beta} \end{bmatrix} = \begin{bmatrix} -\cos \alpha & 0 \\ \frac{\sin \alpha}{\rho} & -1 \\ -\frac{\sin \alpha}{\rho} & 0 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} \quad \text{for } \alpha \in I_1 \text{ with } I_1 = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right]$$

Where ρ is the distance between the center of the robot's wheel axle and the goal position,

Position control: kinematic model

- On the other hand: if $\alpha \in I_2$ with $I_2 = \left(-\pi, -\frac{\pi}{2}\right] \cup \left(\frac{\pi}{2}, \pi\right]$

Redefining the forward direction of the robot by setting $v = -v$, we obtain a system described by a matrix equation of the form

$$\begin{bmatrix} \dot{\rho} \\ \dot{\alpha} \\ \dot{\beta} \end{bmatrix} = \begin{bmatrix} \cos \alpha & 0 \\ -\frac{\sin \alpha}{\rho} & 1 \\ \frac{\sin \alpha}{\rho} & 0 \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix}$$

A differential Drive Robot: Summary

For a differential-drive robot, the kinematic model is described by the following equations:

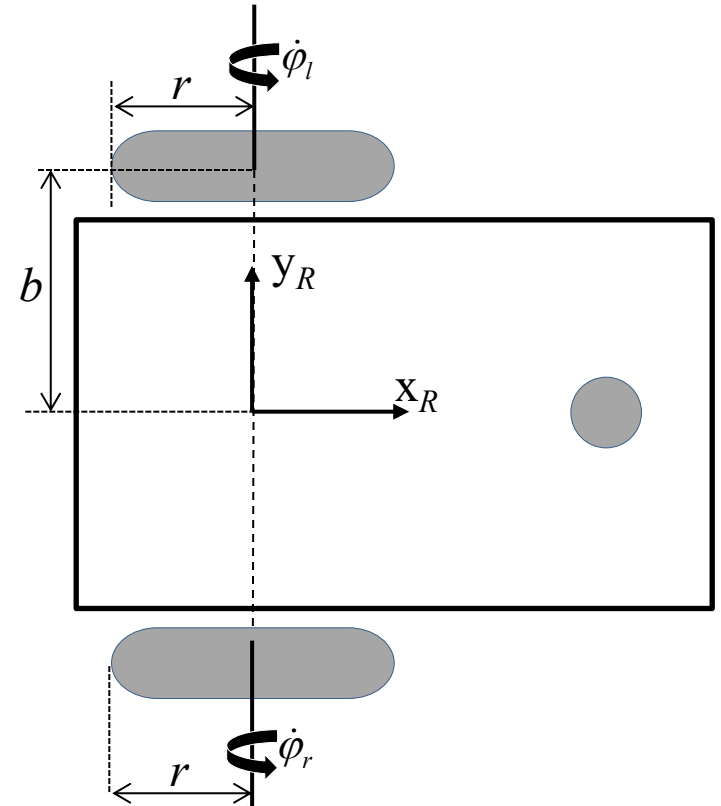
- Forward differential kinematics:

$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}_R = \begin{bmatrix} r/2 & r/2 \\ 0 & 0 \\ r/2b & -r/2b \end{bmatrix} \begin{bmatrix} \dot{\phi}_r \\ \dot{\phi}_l \end{bmatrix}$$

- Inverse differential kinematics :

$$\begin{bmatrix} \dot{\phi}_r \\ \dot{\phi}_l \end{bmatrix} = \begin{bmatrix} 1/r & 0 & b/r \\ 1/r & 0 & -b/r \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}_R$$

$$v = \dot{x} \quad \text{and} \quad \omega = \dot{\theta}$$



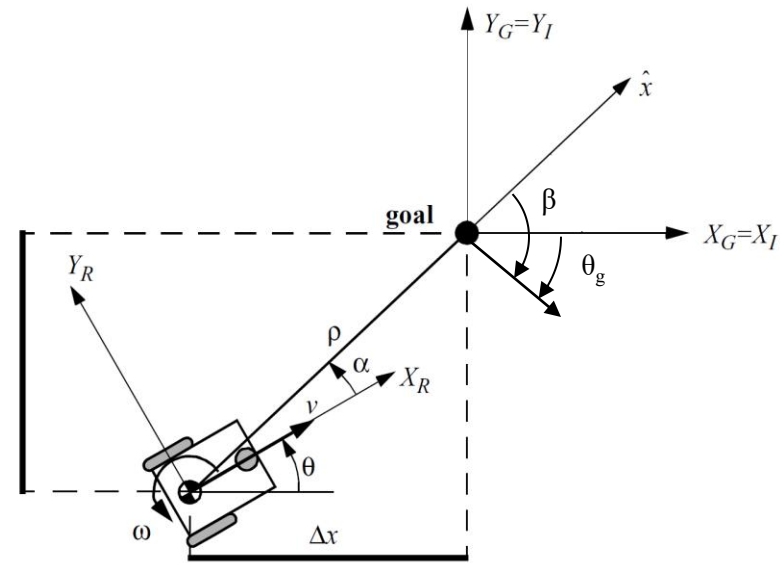
where (v, ω) represent forward and rotational velocity of the robot platform, respectively, and $(\dot{\phi}_r, \dot{\phi}_l)$ the spinning speed of the right and left wheels. The wheel radius is given by r and b denotes half of the inter-wheel distance.

Position Control: Remarks

- The coordinates transformation is not defined at $x = y = 0$;

For $\alpha \in I_1$ the forward direction of the robot points toward the goal,
for $\alpha \in I_2$ it is the backward direction.

$$\alpha \in I_1 = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right]$$



- By properly defining the forward direction of the robot at its initial configuration, it is always possible to have $\alpha \in I_1$ at $t=0$
- However this does not mean that α remains in I_1 for all the time

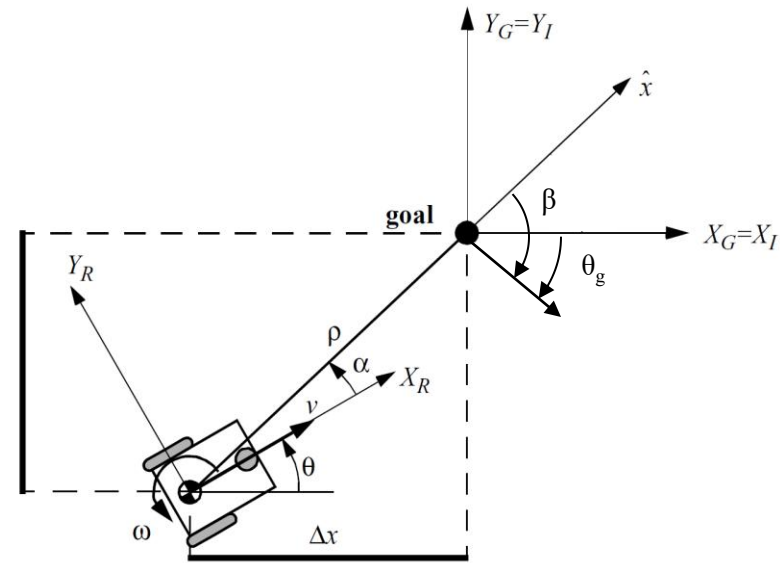
Position Control: control law

- It can be shown, that with $v = k_\rho \rho$ $\omega = k_\alpha \alpha + k_\beta \beta$

the feedback controlled system

$$\begin{bmatrix} \dot{\rho} \\ \dot{\alpha} \\ \dot{\beta} \end{bmatrix} = \begin{bmatrix} -k_\rho \rho \cos \alpha \\ k_\rho \sin \alpha - k_\alpha \alpha - k_\beta \beta \\ -k_\rho \sin \alpha \end{bmatrix}$$

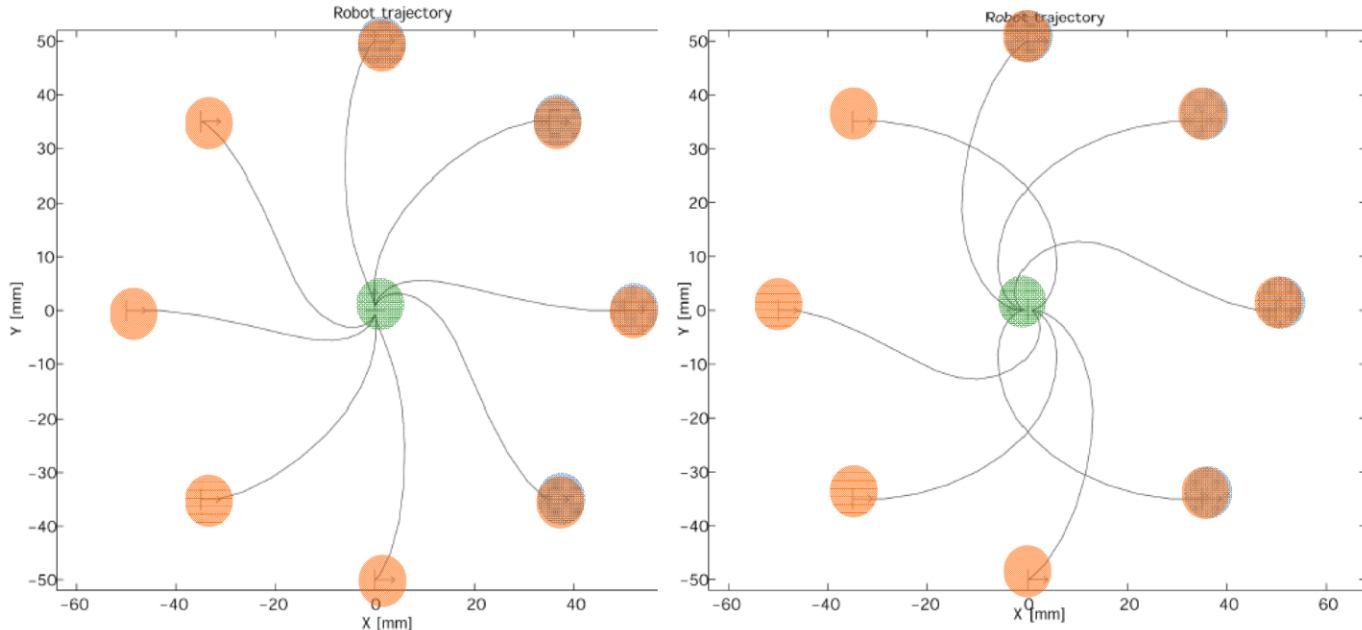
will drive the robot to $(\rho, \alpha, \beta) = (0, 0, 0)$



- The control signal v has always constant sign
 - The direction of movement is kept positive or negative during movement
 - Parking maneuver is performed always in the most natural way and without ever inverting its motion

Position control: Resulting Path

The goal is in the center and the initial position on the circle.



$$\mathbf{k} = (k_p, k_\alpha, k_\beta) = (3, 8, -1.5)$$

For robust position control, it might be advisable to apply the strong stability condition, which ensures that the robot does not change direction during its approach to the goal:

$$k_p > 0 ; \quad k_\beta < 0 ; \quad k_\alpha + \frac{5}{3}k_\beta - \frac{2}{\pi}k_p > 0$$

This implies that $\alpha \in I_1$ for all t , whenever $\alpha(0) \in I_1$ and $\alpha \in I_2$ for all t , whenever $\alpha(0) \in I_2$ respectively. This strong stability condition has also been verified in applications.